Detim Zhao

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Education

Bachelor of Science, Software Engineering May 2026
Arizona State University, Polytechnic (GPA: 3.95 / 4.00) *Mesa, AZ*

- Barrett Honors College, Dean's List (Fall 2022, Spring 2023, Fall 2023, Spring 2024, Spring 2025)

Technical Skills

Languages: Java, Python, C, C++, SQL (PostgreSQL/MySQL), Bash/Shell Scripting
Technologies: pandas, scikit-learn, matplotlib, seaborn, NumPy, PyTorch, Transformers (Hugging Face), OpenCV
Tools: Docker, AWS, Git/GitHub, Unreal Engine 5 (UE5), LaTeX, Jupyter Notebook, UML/PlantUML/Astah
Methodologies: Agile/Scrum, Git Workflows

Experience

Extended Reality (XR) Creative Developer July 2024 – Present
Meteor Studio *Mesa, AZ*

- Containerized GPU-enabled multi-dependency environment using Docker Compose, reducing setup time by 73% (60 to 16 minutes) and eliminating Linux/WSL2 compatibility issues.
- Built volumetric rendering pipeline in Python using RANSAC and ICP algorithms for precise 3D camera calibration in research prototype applications.
- Developed UE5 prototype of autonomous stereoscopic-camera drone with runner tracking and custom control UI.
- Co-authored two research posters on volumetric rendering and camera calibration for ACM HotMobile 2025.

Undergraduate Researcher June 2023 – August 2023
Texas A&M-Corpus Christi *Corpus Christi, TX*

- Developed multi-drone control system using Python with fellow researcher for PX4 drones in Gazebo simulation, coordinating up to 8-10 autonomous drones via terminal commands.
- Preprocessed two datasets using Python, then trained and tested five ML models (KNN, SVM, Random Forest, Naive Bayes, Ensemble) for gesture recognition, enhancing classification accuracy by 10%.
- Co-authored research published in ACM Digital Library and ICICT 2024.

Projects

ChatGPT-Integrated Emergency Alert System (In Progress)

- Co-designed a containerized, event-driven architecture outlining data ingestion, processing, and retrieval layers connected to ChatGPT through secure endpoints for real-time emergency data access.
- Built a Python ETL service to extract and normalize live alert data from an AWS API Gateway endpoint, preparing structured JSON for downstream processing.
- Planned implementation of a Retrieval-Augmented Generation (RAG) workflow with vector database integration to enable dynamic, context-aware responses.

gRPC Microservices System

- Architected a distributed system with 5 gRPC-based microservices, implementing both interactive CLI and automated client interfaces with robust error handling and modular service design.
- Designed gRPC services with Protocol Buffers to enable efficient serialization, persistent data storage, and flexible request handling across multiple microservices.
- Deployed system on AWS EC2 instance with public accessibility, enabling multi-client access and validating concurrent service requests.

Clustering Analysis of Steam Games Using Reviews, Sentiments, and Metadata

- Conducted clustering analysis with Python on 42,497 Steam games using K-Means and Agglomerative Clustering with custom distance metrics, documenting process and findings in academic paper format.
- Scraped and analyzed the sentiment of 385 games' English reviews using VADER and vectorized text with BERT embeddings, transforming Steamworks API JSON data into structured formats for clustering and analysis.
- Built a modular Python pipeline for data collection, preprocessing, and ML analysis using pandas, scikit-learn, and transformers, managed with Git/Git LFS.